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Response to Technical Check

Manuscript Title:

OmniGene-4: A Unified Bio-Language MoE Model with Router-Level Interpretability

Manuscript ID: 45376cf0-cd38-4848-9c46-c01e0dff0a15 v1.0

Journal: *Communications AI & Computing*

Dear Swetha M and the Editorial Team,

Thank you for the technical check of our submission. We have addressed each of the three requested items and re-uploaded the revised manuscript. Below we describe the specific changes made.

Editor’s request 1: Funding Statement

“The Funding Statement is not present in the manuscript. Please provide a Funding Declaration in the manuscript that includes the name of the funding source and the grant number, as a separate sub heading. If there was no Funding, please state so. If no grant number is applicable, kindly indicate this as ‘Not applicable.’ Also make sure to maintain consistency with the submission system funding statement given.”

Author response:

A new *Funding* section has been added to the manuscript as a separate sub-heading, immediately before the *Acknowledgments* section. The exact text reads:

Funding

This work received no specific grant funding from any agency in the public, commercial, or not-for-profit sectors. Grant number: Not applicable. GPU compute resources used for model training were donated by io.net (see Acknowledgments).

This statement is consistent with the funding declaration provided in the submission system (no specific grant funding; GPU compute donated by io.net acknowledged separately).

Editor’s request 2: Figure 4 in-text citation

“Figure 4 is not cited in the main text. Please ensure that all figures are appropriately cited in the manuscript to maintain clarity and consistency.”

Author response:

Figure 4 (the head-to-head accuracy matrix on the five case-study protein pairs) is now explicitly cited in Section 3.5 (*Case Studies and Comparison Against Classical Baselines*). The new reference sentence reads:

“... (Type D, 1 case) label 1, all methods including v5 wrong (honest failure mode). Figure 4

shows the per-method correctness pattern across these five pairs as a colour-coded matrix (blue: correct, red: wrong).”

Editor’s request 3: Tables 4, 5, and 8 in-text citations

“Table 4, 5 and 8 is not cited in the main text. Please include an in-text citation for Table 4, 5 and 8 to ensure all tables are properly referenced and integrated into the manuscript.”

Author response:

In-text citations have been added for all three tables.

- **Table 4** (*Stage-wise decomposition of cross-task routing differentiation*, the CPT-vs-SFT 96%/4% table) is now cited in Section 3.3.3. The lead-in sentence has been extended to read:

“... for the mean pair-wise JS at layer ℓ , we obtain the stage-wise decomposition shown in Table 4.”

- **Table 5** (*Per-task average JS-to-other-tasks*) is now cited in Section 3.3, immediately before the table. The new sentence reads:

“A complementary view is per-task isolation: for each task we compute its average JS divergence to the other seven tasks, layer-averaged; the result is summarized in Table 5.”

- **Table 8** (*CPT corpus composition*) is now cited in Section M.2 (Continued pretraining) of the Methods. A short lead-in sentence has been added before the table:

“The CPT corpus is a 32.5 GB sampled mixture of DNA, protein, natural language, and structural sources, with composition summarized in Table 8.”

Verification.

After the changes above, we re-checked every figure and table label in the manuscript. All 15 display items (4 figures, 11 tables) now have at least one in-text citation. The revised manuscript compiles cleanly to 25 pages with no undefined references and no formatting errors. No other content has been modified.

We thank the editor and the technical check team for their careful review and remain available for any further clarification.

Sincerely,

Liang Wang